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Students' Perceptions of Online Tools in CAD Education

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ABSTRACT

This paper presents results from two web-based surveys (start-up and mid-term) executed during a first year CAD course. As a result, students' opinions about how different online tools support their learning and student's willingness to use rather technical CAD programs at home were studied. This information will help to focus and utilize teaching resources better and enables to move the teaching of CAD from learning the software to utilizing computer-aided tools and methods in a product design process.

Conference Key Areas: Engineering Skills, Open and Online Engineering Education, Curriculum Development

Keywords: computer aided design, blended learning, mechanical engineering

INTRODUCTION

In the field of mechanical engineering, computer aided tools are part of everyday life. Increased quality standards, new production methods and faster pace of product development drives the education at the universities. Traditionally, teaching of Computer Aided Design (CAD) relies on computer class exercises in fixed places and predefined times involving numerous teaching assistants to deal with questions and problems of the students. This approach has worked well during the years, when software was only installed in local computers using local licenses. Nowadays, based on the survey done for students in the freshman year CAD course, 94% of students have a laptop computer, 38% have a tablet computer and 44% have a desktop computer. The licensing policy and increased sharing of software encourage

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students to complete exercises whenever and wherever they choose. Developments in the web-based learning environments makes sharing and receiving exercises more efficient and accessible. This allows utilizing blended learning approach, which has provided good evidence to support learning in higher education [1]. In blended learning, both traditional classroom interaction and online learning are combined in order to increase "*pedagogical richness, access to knowledge, social interaction, personal agency, cost effectiveness, and ease of revision*" [2].

A study done by Hannay and Newvine [3] suggests that students believe they learn more, spend more time on studying and found additional instructor materials more useful in distance learning than in contact learning. Distance learning, like e-Learning and online learning, is a term describing learning that is done somewhere else than at university. The main difference between these terms is in the form of implementation.[4] Combining online elements with traditional classroom teaching can enrich students learning environment. To better support blended learning approach in CAD teaching, it is essential to systematically collect student's perceptions of online tools. Thus, research questions for this paper are:

- What kind of online materials are suitable for CAD teaching?
- What kind of tools/methods students prefer to use?
- Will the students utilize the online tools provided to do exercises time and place independent?

This information will help us to utilize the teaching resources better and the focus of teaching of CAD can be transferred from learning the software to utilizing computer-aided tools and methods in a product design process, i.e. from declarative to procedural knowledge [5].

1 METHODS

The feedback from a freshman CAD course was utilized in order to gain student's perspectives of online tools. *Computer-aided Tools in Engineering* is a 5 ECTS, two period (2x7 weeks) course introducing CAD modelling to engineering students. This course is a part of the bachelor's degree programme and it is an obligatory 1st year course to all students in the School of Engineering. Annually about 300 students complete this course.

Learning outcomes of the course are

- understanding the basics of computer aided tools
- ability to use computer aided tools.

The course is divided into two modules (*Fig. 1*): the common module, which all students take during the first teaching period (7 weeks), and the elective module (6 weeks), where students choose the most interesting one from the offered software modules.

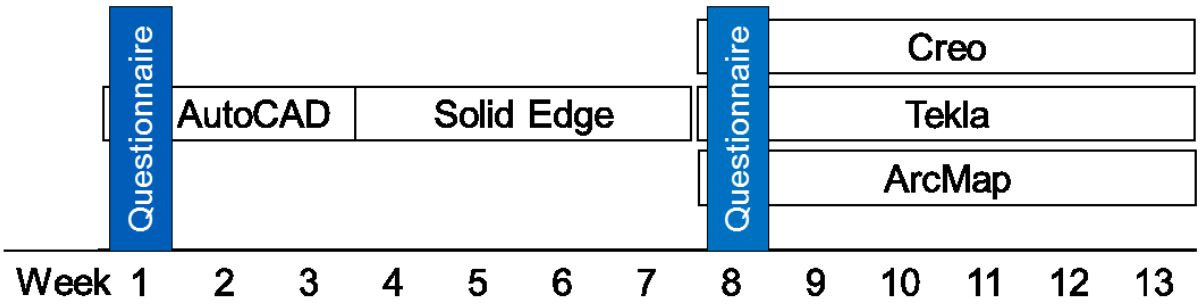


Fig. 1. Structure of the course

In the common module, basics of 2D and 3D modelling are introduced. Students start with two-dimensional drawing using Autodesk AutoCAD (3 weeks, 2 exercises) continuing to three-dimensional modelling using Siemens Solid Edge (4 weeks, 5 exercises). In this way, all students in the school become familiar with general tools used in both civil and mechanical engineering. The module contains lectures related to usage of the software and their general application fields. These skills are applied on the weekly supervised exercises on the computer room.

In the elective part, students choose a software module based on their interest. In the spring 2017, elective applications were PTC Creo for students interested in mechanical engineering, Tekla Structures for students interested in civil engineering and Esir ArcMap for students interested in surveying and environmental engineering.

The exercises during the 3D CAD common module are given and supported by various online materials and methods:

- Video without sound showing step-by-step how exercise is completed.
- Video without sound demonstrating how a certain tool works.
- Video with narrative demonstrating how a certain tool works.
- Viewable model in browser-based CAD of exercise specific object.
- PDF material showing step-by-step how exercise is completed.
- PDF material listing hints and suitable tools needed for the exercise.
- PDF in book format.
- Shared PowerPoint presentation with embedded videos demonstrating how tools work.
- Auto-assessed quiz about engineering drawing topics, standards etc.
- Interactive engineering drawing including explanations of related markings.
- FAQ forum about modelling and software.

This material was distributed using Moodle-based learning environment on the weekly basis or at the start of the module.

For this study, two questionnaires were carried out (*Fig. 1*): a start-up (N=330) and a mid-term questionnaire (N=279). They were all realised using Webropol, a web-based survey tool. Students were informed that results from these questionnaires would be collected and be used to develop teaching in the course and no further instructions were provided.

The start-up questionnaire was organized during the first week of the course. The questions considered the previous experience with computers and different software types (such as office tools, editors and, mobile applications) as well as what kind of devices the students own.

The mid-term questionnaire was arranged after the common module of the course and questions were related to online material.

2 RESULTS

The surveys were composed of mainly closed questions (5 or 10 point scale). Statistics for online surveys were obtained using statistical tools of the survey software (Webropol). The results are presented using the translated descriptors that were used in the questionnaires.

2.1 Software usage skills

On average, students had been using computers for 12.5 years and they estimated that they use computer for 4.5 hours per day. Students were also asked to estimate their software skills with different types of software on a scale from 1 to 10, where 1 is a novice and 10 an experienced user (*Table 1*).

Table 1. Students' software skills as estimated by themselves

Software type	Average skill
Email applications	7.3
Streaming applications	7.3
Office application suites	6.9
Mobile applications	6.7
Social media applications	6.4
Computer games	5.7
Mobile games	5.7
Programming editors	4.9
Mathematical software	4.1
Picture/movie editors	3.6
Activity applications	2.5
Web page editors	2.2
2D CAD	1.9
3D CAD	1.8

2.2 Preferred online tools or material

In the mid-term survey, students were additionally asked to estimate how the different types of course material supported their learning during the common part of the course. This result is presented in *Fig. 2* using diverging stacked bar charts as presented in [6]. The scale was 5-point Likert with "Didn't use" (*Fig. 3*) option.

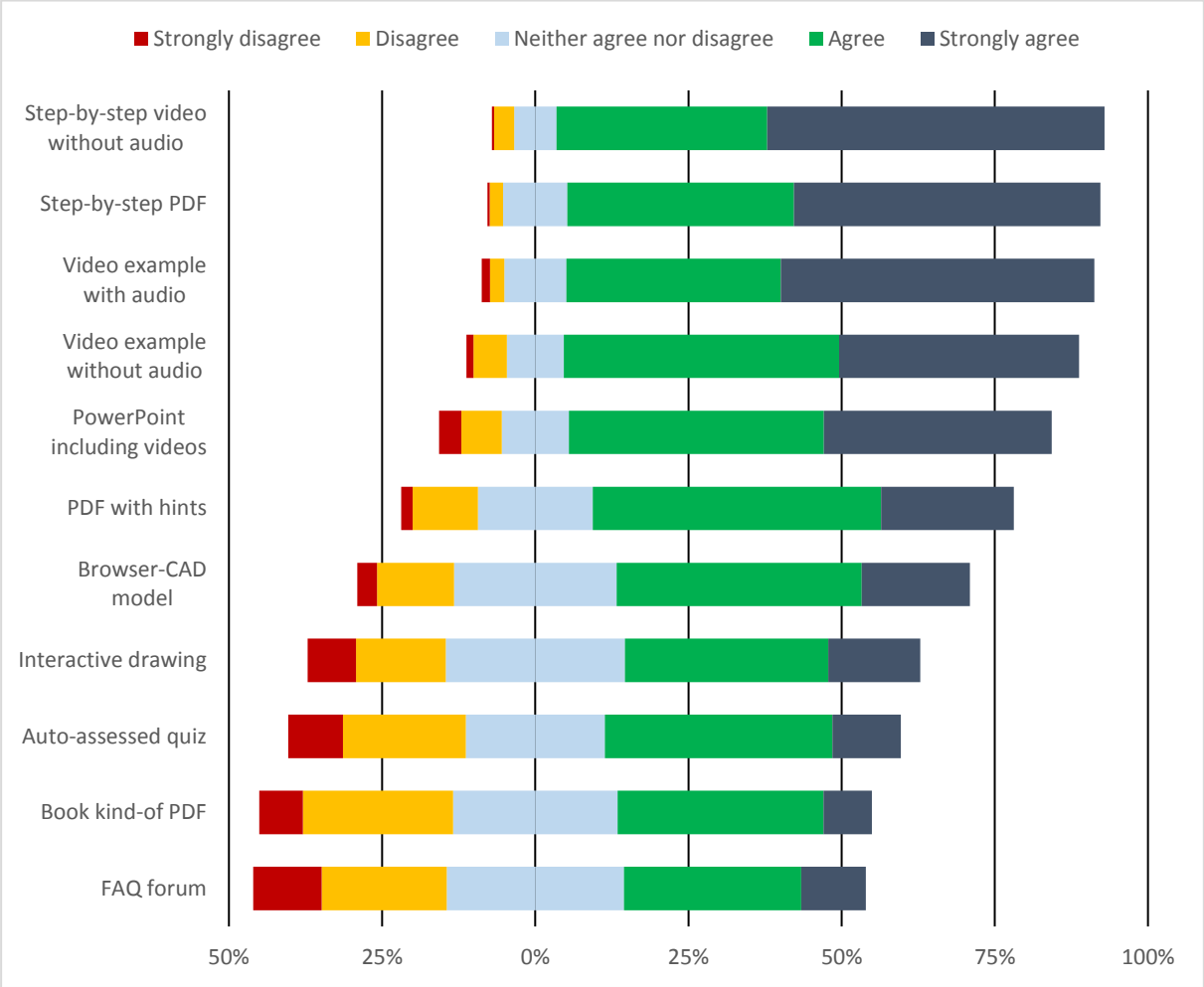


Fig. 2. Students' answers to question: "The following tools/materials supported my learning during 3D CAD part of the course"

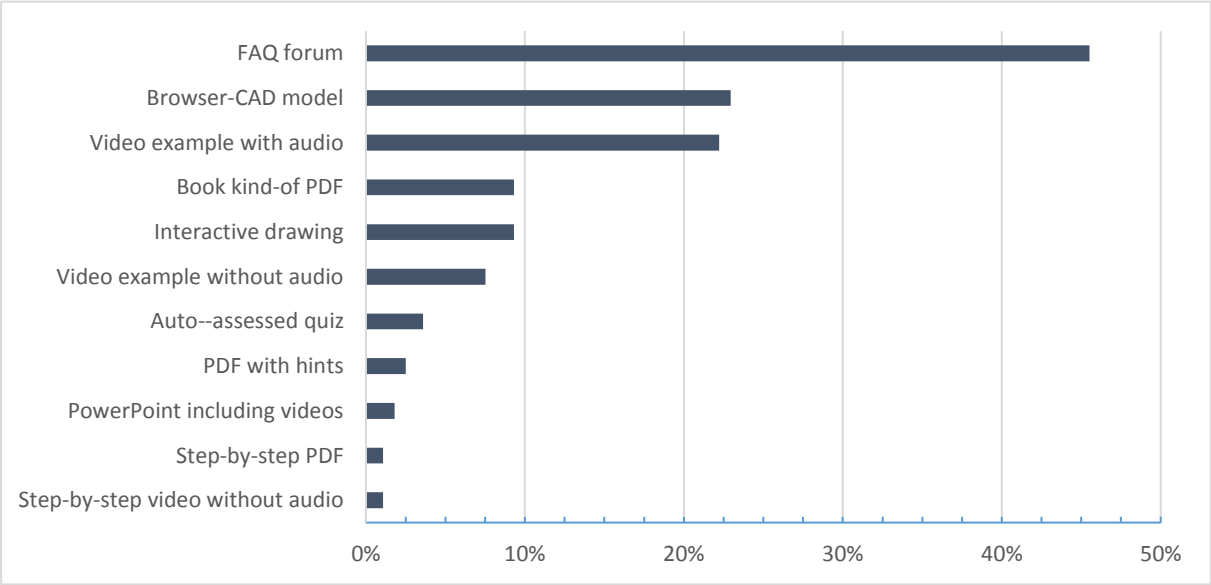


Fig. 3. Percentage of students who didn't use provided tools/materials

2.3 Utilization of provided software

In the mid-term survey students were asked if they had installed the provided 3D CAD software (Siemens Solid Edge) on their home computers (*Table 2*). The

software and licence were distributed using the university's software download portal and no registration was required.

Table 2. Answers to survey questions related to software usage

Did you install 3D CAD software on your home computer?	**%**
Yes	54.8%
No	45.2%
If No, mention max three reasons	%
Insufficient performance	39.7%
Wanted no extra software	38.9%
Incompatible software	21.4%
Did not want to use own computer	15.1%
Found hard to install	15.1%
Incompatible hardware	3.2%

When asked how they carried out the weekly exercises, 69.5% of the students responded having started the task before the actual class exercise and 56.3% had completed it together with other students.

3 DISCUSSION

Students' starting skills with CAD tools were low (1.9 in 2D and 1.8 in 3D), which was expected for the freshman course. Students had good skills in social media (6.4), streaming (7.3), emailing (7.3) and office suites (6.9). Data also shows how confident students are with mobile application and gaming (6.7 and 5.7). These outcomes can be taken into account while creating or selecting study materials or designing courses in the future, i.e. ensuring that material supports mobile devises. The utilization of social media applications in CAD teaching needs to be looked into.

Students seem to prefer teaching materials that show step-by-step how the CAD model is created, both in video and PDF formats. Video materials were among the most preferred material types and presence of narrative in videos did not make a notable difference. This is an interesting finding and may lower the teacher's threshold to produce video materials.

Shared PowerPoint material including embedded videos were also preferred. This material type combines both written and video formats, both which were highly preferred by the students. From teacher's point of view, creating this kind of material is somewhat easier and tools are more familiar (contrary to video editing software). For the next year CAD course, more study in this field is preferred.

Surprisingly, three tested interactive material types (browser-CAD model, interactive drawing and auto-assessed quiz) were not preferred. One of the exercises was supported by both browser-CAD and step-by-step video, which might explain the lower grade of browser-CAD model – 23% of students did not use it at all.

Interactive drawing and auto-assessed quiz formed one of the exercises, in which students practiced the basic principles of 2D engineering drawings. This exercise was not so much about actual 3D modelling, but about documentation phase.

The least liked and used support material type (45.5% did not use it) was the FAQ forum. This forum was created in the course learning environment. It seems that there was no need to use the forum or students got the help from other sources. On the other hand, the usability of the forum may not satisfy the students' expectations.

Almost half of the students (45.2%) did not install the CAD software on their home computer. The main reasons were both technical – lack of performance (39.7% of those who did not install) and incompatible software (21.4%), as well as practical – students did not want to install extra software (38.9%). It seems that computer classes are suitable for students, maybe because they offer a place to study together.

Development of online course support material is a continuous process while the field is constantly changing and students adopt new tools and applications especially in the mobile world.

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REFERENCES

[1] Garrison, D., Kanuka, H., (2004), Blended learning: Uncovering its transformative potential in higher education, *The Internet and Higher Education*, Volume. 7, Issue 2, pp. 95-105.

[2] Osguthorpe, R. T., Graham, C. R., (2003), Blended learning environments: Definitions and directions, *Quarterly Review of Distance Education*, Volume 4, Issue 3, pp. 227-33.

[3] Hannay, M., Newvine, T., (2006), Perceptions of Distance Learning: A Comparison of Online and Traditional Learning, *Journal of Online Learning and Teaching*, Volume 2, Issue 1, pp. 1-11.

[4] Moore, J. L., Dickson-Deane, C., & Galyen, K. (2011), e-Learning, online learning, and distance learning environments: Are they the same?, *The Internet and Higher Education*, Volume 14, Issue 2, pp. 129-135.

[5] Hamade, R. F., Hassan, A. A., Mohamad, Y. J., (2007), Evaluating the learning process of mechanical CAD students, *Computers & Education*, Volume 49, Issue 3, pp. 640-661.
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[6] Robbins, N. B., Heiberger, R. M., (2011), Plotting Likert and Other Rating Scales, Joint Statistical Meetings, Survey Research Methods, pp. 1058-1066.